

Diabetes Mellitus Fundamentals

MBBS Community Medicine Posting Handout

Diabetes Mellitus: Essential Knowledge

Definition

Diabetes mellitus is a chronic metabolic disorder characterized by hyperglycemia resulting from defects in insulin secretion, insulin action, or both.

Prevalence (2023 Data)

- **Global:** 537 million adults affected (IDF Atlas 10th Edition)
- **India:** 77 million people affected (15.3% of adults 20-79 years)
- **Rural India:** 57% of diabetics remain undiagnosed
- **Projection 2030:** India's diabetes burden to reach 93 million

Classification

1. Type 1 DM (5-10%)

- Autoimmune destruction of β -cells
- Absolute insulin deficiency
- Presentation: Young age, ketoacidosis
- Management: Insulin replacement

2. Type 2 DM (90-95%)

- Insulin resistance + progressive β -cell failure
- Strong genetic component
- Presentation: Adult onset, obesity
- Management: Lifestyle + oral agents + insulin

3. Other Types (1-2%)

- Gestational diabetes
- Monogenic diabetes (MODY)
- Secondary diabetes (pancreatitis, drugs)

Diabetes Pathophysiology

Normal Glucose Regulation

- **Insulin:** Promotes glucose uptake and glycogen synthesis
- **Glucagon:** Increases blood glucose by promoting glycogenolysis
- **Counter-regulatory hormones:** Epinephrine, cortisol, growth hormone

Pathogenesis of Type 2 DM

1. **Genetic Predisposition:** Multiple gene defects
2. **Insulin Resistance:** Peripheral tissues fail to respond normally to insulin
3. **β -Cell Dysfunction:** Progressive loss of insulin secretory capacity
4. **Compensatory Hyperinsulinemia:** Initial response, then failure

Hyperglycemia Effects

- **Acute:** Polyuria, polydipsia, fatigue, blurred vision
 - **Chronic:** Microvascular/macrovascular complications
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Clinical Features

Classic Triad

- **Polyuria:** Frequent urination
- **Polydipsia:** Excessive thirst
- **Polyphagia:** Increased hunger (in type 1)

Other Symptoms

- **Fatigue and weakness**
- **Blurring of vision**
- **Recurrent infections**
- **Delayed wound healing**
- **Weight loss** (type 1)
- **Asymptomatic** in 50% cases (type 2)

Complications Screening (Annual)

- **Retinopathy:** Fundoscopy or dilated eye examination
 - **Nephropathy:** Urine albumin/creatinine ratio, eGFR
 - **Neuropathy:** Foot examination, vibration sense, reflexes
 - **Cardiovascular:** ECG, lipid profile, blood pressure
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Diagnostic Criteria (ADA 2023)

Fasting Plasma Glucose (FPG)

- **Normal:** < 100 mg/dL (5.6 mmol/L)
- **Pre-diabetes:** 100-125 mg/dL (5.6-6.9 mmol/L)
- **Diabetes:** $\geq$ 126 mg/dL (7.0 mmol/L) on two occasions

2-Hour Oral Glucose Tolerance Test (OGTT)

- **Normal:** < 140 mg/dL (7.8 mmol/L)
- **Pre-diabetes:** 140-199 mg/dL (7.8-11.0 mmol/L)

- **Diabetes:** ≥ 200 mg/dL (11.1 mmol/L)

HbA1c

- **Normal:** $< 5.7\%$ (39 mmol/mol)
- **Pre-diabetes:** 5.7-6.4% (39-47 mmol/mol)
- **Diabetes:** $\geq 6.5\%$ (48 mmol/mol)

Random Blood Glucose

- **Diabetes:** ≥ 200 mg/dL (11.1 mmol/L) + symptoms
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Risk Factors

Non-Modifiable

- **Age ≥ 35 years**
- **Family history** (first-degree relatives)
- **Previous gestational diabetes**
- **Ethnicity** (Asian Indians – 2-4 times higher risk)

Modifiable

- **Obesity/BMI ≥ 25 kg/m²**
 - **Abdominal obesity** (waist ≥ 90 cm in men, ≥ 80 cm in women)
 - **Physical inactivity** (< 150 minutes moderate activity/week)
 - **Unhealthy diet** (high refined carbs, saturated fats)
 - **Tobacco use** and excessive alcohol
 - **Hypertension** (BP $\geq 140/90$ mmHg)
 - **Dyslipidemia** (low HDL, high triglycerides)
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Management Principles

General Approaches

1. **Medical Nutrition Therapy (MNT)**
2. **Physical Activity & Exercise**
3. **Pharmacological Therapy**
4. **Self-Monitoring of Blood Glucose (SMBG)**
5. **Education & Psychological Support**
6. **Regular Physician Follow-up**

Treatment Goals (ADA 2023)

Parameter	Target
HbA1c	$< 7.0\%$ ($< 6.5\%$ if young, no contraindications)
Fasting BG	80-130 mg/dL (4.4-7.2 mmol/L)

Parameter	Target
Postprandial BG	< 180 mg/dL (< 160 mg/dL after 2 hours)
Blood Pressure	< 130/80 mmHg
LDL Cholesterol	< 100 mg/dL (< 70 mg/dL if CVD)

Pharmacological Treatment

Type 2 DM Treatment Algorithm

1. **Metformin** (first-line): 500-1000 mg twice daily
2. **Add second agent:** DPP-4 inhibitors (sitagliptin, teneligliptin)
3. **Triple therapy:** Add SGLT2 inhibitors or sulfonylureas
4. **Insulin initiation:** When optimal dual/triple therapy fails

Common Oral Agents

Class	Examples	Mechanism	Side Effects
Biguanides	Metformin	Decrease hepatic glucose output	GI upset, lactic acidosis
Sulfonylureas	Glimepiride	Stimulate insulin secretion	Hypoglycemia, weight gain
DPP-4 Inhibitors	Sitagliptin	Increase incretin levels	Rare - pancreatitis
SGLT2 Inhibitors	Empagliflozin	Block glucose reabsorption	UTI, genital infections
GLP-1 Agonists	Dulaglutide	Mimic incretin effects	Nausea, pancreatitis

Insulin Therapy

- **Basal/Bolus regimen:** Long-acting (e.g., insulin glargine) + short-acting
- **Starting dose:** 0.3-0.5 IU/kg/day
- **Adjustment:** Based on SMBG readings
- **Education essential:** Dosage, administration, storage

Diabetes Complications

Acute Complications

1. **Hypoglycemia:** Common with insulin/sulfonylureas
 - Symptoms: Sweating, tremor, confusion, coma
 - Treatment: 15g carbs, glucagon, hospitalization if severe
2. **Hyperglycemic emergencies:**
 - **Diabetic Ketoacidosis (DKA):** Type 1, infection, miss insulin
 - **Hyperosmolar Hyperglycemic State (HHS):** Type 2, severe dehydration

- Treatment: Fluid resuscitation, insulin, electrolyte correction

Chronic Complications

Macrovascular (Cardiovascular)

- Coronary artery disease (2-4x risk)
- Cerebral vascular disease (stroke 2-4x risk)
- Peripheral vascular disease (leading cause of amputation)

Microvascular (Classic Triad)

1. **Diabetic Retinopathy:** Leading cause of blindness
 - Non-proliferative → proliferative → maculopathy
 - Screening: Annual dilated fundoscopy from diagnosis
 2. **Diabetic Nephropathy:** Leading cause of ESRD
 - Microalbuminuria → macroalbuminuria → CKD
 - Screening: Annual UAE and serum creatinine/eGFR
 3. **Diabetic Neuropathy:** Most common complication
 - Distal symmetric polyneuropathy
 - Autonomic neuropathy (gastroparesis, orthostatic hypotension)
 - Screening: Foot examination, monofilament testing
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Community Medicine Perspective

Prevention Strategies

- **Population Level:** WHO "Best Buys" - sugar taxation, food labeling
- **High-Risk Individuals:** Lifestyle interventions (DPP trials)
- **Screening Programs:** NPCDCS guidelines for community screening
- **Health Promotion:** Community awareness through schools, workplaces

Government Programs in India

- **NPCDCS (2010):** National Programme for Prevention & Control
- **NP-IHP (2017):** Non-Communicable Diseases component
- **Free Medications:** Oral hypoglycemics, insulin through government schemes
- **PP-Hypertension & Diabetes:** Prevention & control program

Role of ASHA in Diabetes Care

- Community screening and awareness
- Follow-up of registered diabetics
- Referral coordination
- Community mobilization for prevention

Key Clinical Skills

Community Screening

- Consent procedures and privacy
- Glucometer operation and result interpretation
- Anthropometric measurements (BMI, waist circumference)
- Health education and counseling

Patient Management

- Individualized education approach
- Family involvement strategies
- Cultural sensitivity in counseling
- Referral system navigation

Prevention Focus

- Lifestyle modification counseling
- Community health promotion
- School and workplace interventions
- Research and data analysis

References for Further Reading

1. **ADA Guidelines 2023:** American Diabetes Association Standards of Care
2. **WHO PEN Protocol:** Package of Essential Non-Communicable Diseases Interventions
3. **NCDIR Guidelines:** National Guidelines for Management of Diabetes in India
4. **IDF Clinical Practice Recommendations**
5. **RSSDI Textbook of Diabetes Mellitus**

Key Takeaways for Clinical Practice

1. **Screening is essential** - Early diagnosis prevents complications
2. **Lifestyle modification** is the cornerstone of management
3. **Individualization** of therapy is crucial for success
4. **Team approach** improves patient outcomes
5. **Regular monitoring** prevents acute and chronic complications
6. **Patient education** enhances self-management capabilities

Note: This handout serves as a quick reference guide. Students should refer to standard textbooks and recent guidelines for comprehensive understanding.

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